Patrick Bahati

Data Scientist | AI & Automation | SAP ABAP



CONTACT

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PERSONAL INFORMATION

Nationality: Rwandan
Date of birth: 25 March 1999

LANGUAGES

Mother tongue: Kinyarwanda

English	C1
Luganda	C1
Swahili	B1
Italian	B1

CEFR self-assessed; full skill breakdown in Language Skills section.

DIGITAL SKILLS

Information & data literacy – Proficient
Communication & collaboration – Proficient
Content creation – Proficient
Safety – Independent
Problem solving – Proficient

KEY SKILLS

- Programming
- SAP Development
- Machine Learning
- Federated Learning
- Big Data
- Scientific Computing
- Databases
- Data Visualization
- Computer Vision
- AI & Automation
- Web & Backend
- DevOps & Tools
- Research Methods

ABOUT ME

I am a **Data Scientist and AI Automation Developer** with hands-on experience in machine learning, deep learning, big data engineering, AI-powered business automation, and modern SAP development. I am currently completing a **Master's Degree in Data Science** at the **Università degli Studi di Messina**, with research experience in spiking neural networks and computational neuroscience. I have practical training in **AI & Business Automation** and modern **SAP ABAP Cloud development**, including SAP BTP, CDS, RAP, OData V4, and Clean Core principles. I am comfortable working across technical and business environments and communicating with both technical and non-technical stakeholders.

WORK EXPERIENCE

Computational Neuroscience Research Intern — *March 2026 – June 2026*
Università degli Studi di Messina — Brian Lab — *Messina, Italy*

Sector: Higher education / scientific research

- Implemented biologically inspired Spiking Neural Networks (SNNs) using the Brian2 simulator.
- Developed and simulated Leaky Integrate-and-Fire (LIF) and conductance-based neuron models.
- Designed STDP and Reward-Modulated STDP (R-STDP) learning systems.
- Investigated excitatory/inhibitory synaptic dynamics and eligibility trace mechanisms.
- Visualized membrane voltage, synaptic conductance, and adaptive weight changes including LTP and LTD.
- Built end-to-end experimental SNN simulation pipelines for computational neuroscience research.

Business Coach — *October 2023 – October 2024*
AVSI Foundation — *Isingiro District, Nakivale, Uganda*

Sector: International development / NGO

- Mentored micro and large enterprises using data-informed business growth strategies and market analysis.
- Collaborated with private-sector employers on job placement, reporting labor market data for workforce development.

Data Analyst Intern — *June 2022 – September 2022*
Stratcom Communications and Computer Solutions — *Uganda*

Sector: ICT services

- Cleaned, processed, and analyzed datasets to support data-driven business decisions.
- Developed dashboards and automated reporting solutions using Python and BI tools.
- Supported system performance monitoring and optimization through real-time data analysis.

Data Clerk & Research Assistant — *2021 – 2022*
Windle International Uganda — *Uganda*

Sector: Education / NGO (refugee education)

- Managed large-scale field data collection, validation, and database maintenance for education-focused research.
- Performed quantitative and qualitative analysis while ensuring data accuracy and research compliance.

Community Facilitator — *2019 – 2020*
DRDIP — *Isingiro District, Uganda*

Sector: Community / rural development

- Facilitated stakeholder–community communication and delivered capacity-building programs with outcome monitoring.

EDUCATION AND TRAINING

Master's Degree in Data Science *2024 – Present*
Università degli Studi di Messina *Messina, Italy*

Supervised Machine Learning: Regression and Classification *October 2026*
DeepLearning.AI & Stanford Online (Andrew Ng) — Coursera

- Completed the first course of the Machine Learning Specialization: linear and logistic regression, gradient descent, regularization, and classification.
- Verified at coursera.org/verify/Q2VYSO6M7Q6N.

AI & Business Automation — ReadyForIT (Refugees & Migrants) *September 2026*
Fondazione Italiana Accenture ETS & The Human Safety Net, delivered by Academy Rapido *Rome, Italy*

- Built AI-powered business automation workflows using n8n, OpenAI, Google Gemini, REST APIs, and Google Workspace.
- Applied prompt engineering, RAG pipelines, AI agents, and multi-agent workflow design to business automation projects.

SAP ABAP Cloud Development *2026*
Training & Practical Projects

- Completed practical training in modern SAP development using ABAP Cloud, SAP BTP, CDS Views, RAP, OData V4, and Clean Core.
- Built sample ABAP applications, semantic CDS Views, and a RAP Business Object with service exposure via OData V4.

Next.js App Router Fundamentals *August 2025*
Vercel *Certificate of Completion*

Bachelor's Degree in Business Information Technology *2020 – 2023*
Nkumba University *Entebbe, Uganda*

Final grade: CGPA 4.45 / 5.00

LANGUAGE SKILLS

Mother tongue(s): Kinyarwanda

Other language(s)

	Understand.		Speaking		Writing
	Listen.	*Read.*	*Interact.*	*Product.*	
English	C1	C1	C1	C1	C1
Luganda	C1	C1	C1	C1	C1
Swahili	B1	B1	B1	B1	B1
Italian	B1	B1	B1	B1	B1

Levels: A1/A2 Basic user – B1/B2 Independent user – C1/C2 Proficient user. Common European Framework of Reference for Languages. *(Self-assessed estimate – adjust individual skills if your actual level differs.)*

JOB-RELATED SKILLS

Programming	Python, SQL, Java, JavaScript, HTML/CSS
SAP Development	SAP ABAP Cloud, SAP BTP, CDS Views, RAP, OData V4, Clean Core, Fiori concepts
Machine Learning	Supervised learning (regression & classification), Scikit-learn, TensorFlow, Keras, PyTorch, Spiking Neural Networks (Brian2), STDP, R-STDP
Federated Learning	Flower, non-IID data partitioning, FedAvg, FedMedian, Byzantine fault tolerance
Scientific Computing	NumPy, Pandas, SciPy, Simulation Modeling, Feature Engineering
Big Data	Spark (PySpark, Spark SQL), Kafka, Hadoop (HDFS, MapReduce), HBase, Hive
Databases	MySQL, PostgreSQL, MongoDB, SQL Server, Oracle
Data Visualization	Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau, Excel
Computer Vision	OpenCV, CNNs for Image Classification
AI & Automation	n8n, OpenAI, Gemini, RAG, AI Agents, Multi-Agent Workflows, Prompt Engineering, REST APIs
Web & Backend	React, Angular, Node.js, Next.js, Spring Boot, ASP.NET Core, Django
DevOps & Tools	Docker, Docker Compose, Kubernetes, Git, GitHub
Research Methods	Data Preprocessing, Statistical Analysis, Academic Writing, Experimental Design

COMMUNICATION SKILLS

Strong cross-cultural communication skills developed through a multilingual background (five languages) and roles that required engaging both technical and non-technical audiences, from presenting computational neuroscience research findings to mentoring non-technical micro-entrepreneurs on business growth strategy at AVSI Foundation.

ORGANISATIONAL / MANAGERIAL SKILLS

Experience independently planning and delivering multi-stage technical projects end-to-end (ML pipelines, SNN simulation studies, SAP development exercises), and coordinating with private-sector employers on workforce development and job-placement reporting during my role at AVSI Foundation.

ADDITIONAL INFORMATION — SELECTED PROJECTS

Federated Learning: Comparing FedAvg vs. FedMedian Aggregation
Python, PyTorch, Flower *GitHub*

- Simulated federated learning with 8 non-IID edge clients on CIFAR-10 using a custom residual CNN, comparing FedAvg and FedMedian aggregation and reaching **82% test accuracy**.
- Designed a Byzantine-fault test (one corrupted client): FedMedian retained **64% accuracy** vs. FedAvg's collapse to **10%**.

Reward-Modulated Spiking Neural Network (R-STDP) *Brian2, Python*

- Designed a conductance-based LIF neural network with excitatory and inhibitory synapse populations.
- Implemented STDP and reward-modulated synaptic plasticity with eligibility traces and visualized learning dynamics.

Plant Disease Detection Using CNN *TensorFlow, Keras, Python*

- Built a CNN-based model for multi-class plant disease classification on the PlantVillage dataset (70,000+ images, 38 classes).
- Applied data augmentation, dropout, batch normalization, and early stopping, achieving ∼**91% test accuracy**.

Real-Time Credit Card Fraud Detection Pipeline *PySpark, Kafka, Hadoop, Docker, Streamlit*

- Architected a Lambda Architecture pipeline processing **280,000+ transactions** using Spark Structured Streaming and Kafka (**sub-2-second latency**) with Hadoop MapReduce batch analytics.
- Trained Random Forest and Logistic Regression models (Spark MLlib) served through a containerized Streamlit dashboard.

Big Data Analytics Final Project — Apache HBase & FDA CAERS Dataset
Hadoop, HBase, Docker, Python *University Project*

- Deployed a multi-node Hadoop/HBase cluster (HDFS, YARN, ZooKeeper) with

Docker and built a Python ETL pipeline converting **149,945** FDA adverse-event reports (**233,821** product records) into HBase via MapReduce ImportTsv.
- Designed a normalized HBase schema (namespaces, column families, versioning, compression) and implemented DDL, CRUD/DML, and cluster administration via the HBase Shell.

AI Customer Support Automation Platform *n8n, OpenAI, Supabase, Airtable, Zendesk* *GitHub*

- Designed an AI support workflow with semantic search, ticket classification, Airtable logging, and Zendesk escalation.
- Built a RAG chatbot using Supabase Vector Store and HTML/CSS analytics dashboards.

Multi-Agent Marketing Campaign Generator *n8n, Google Gemini, Google Docs* *GitHub*

- Developed a multi-agent AI system for campaign planning, content generation, and quality review.
- Automated brand strategy and editorial plan generation with consistent tone enforcement through Google Docs integration.

I hereby authorize the processing of my personal data in accordance with the GDPR (EU Regulation 2016/679).

Date: November 2, 2026

Signature: 